Radiology AI | Frontend product requirements and user journeys

Frontend specification package - October 2026 | Planning baseline, not a repository audit

Product positioning

Purpose - Imaging workflow and clinician-reviewed radiology reporting

Primary personas - Specialist clinician, technician or clinical assistant, administrator, and read-only authorized reviewer.

MVP boundary - Clinical workflow management and physician-reviewed reporting; not an autonomous diagnosis product.

Acceptance - No clinical report becomes signed through AI alone; authorization and backend status controls apply to every action.

Screens and modules

Core screens - Login and institution selection; dashboard; study worklist; patient summary; study details; DICOM viewer; report composer; AI suggestions review; notifications; admin/roles; audit timeline.

Radiology worklist - Filter by modality (X-ray, CT, MRI), status, date, priority and assigned radiologist; sortable columns, saved views, keyboard-first navigation, clear unread/urgent states.

Imaging workspace - Prefer OHIF Viewer with DICOMweb/PACS; display series thumbnails, window/level, zoom/pan, measurements and study metadata where supported. Keep viewer customization isolated behind an adapter.

Reporting workspace - Side-by-side image viewer and structured report editor; findings/impression sections; templates; autosave with conflict handling; draft/review/sign controls; amendments with version history.

AI review - Proposed findings appear in a clearly separated unverified panel with evidence references, confidence caveats and explicit clinician accept/edit/reject; no silent report signing.

Mobile - Study worklist, case notifications, report status and limited approved review only. Do not promise diagnostic image interpretation on a phone.

Representative journeys

Scenario A - radiologist - Open assigned CT study -> verify patient/study identity -> inspect images -> compose findings -> optionally request AI draft -> review evidence -> edit -> submit/sign with authorization.

Scenario B - technician - Register or inspect study metadata -> check ingestion and PACS link -> resolve failed imports -> route study to authorized radiologist; no diagnostic report signing.

Scenario C - urgent case - Urgent worklist item shows priority and source -> radiologist acknowledges -> reviews images -> records findings -> follows institution-defined escalation workflow.